Earth’s development: first, the birth of the planet with the geosphere, in which only inanimate matter existed; secondly, the emergence of life in the biosphere; and finally the epoch in which human activity changed the planet forever—the noosphere.

Sphere interactions

Scientists believe the biosphere has constantly changed. Oxygen levels in the atmosphere began to rise at least 2.7 billion years ago, as microorganisms called cyanobacteria multiplied. As oxygen increased, more complex life forms evolved that would shape Earth in different ways, eroding and remolding its surface, and changing its chemical composition.

Gradually, over time, elements of the biosphere became part of the lithosphere. Over millennia, dead corals created reefs in shallow tropical oceans. Similarly, the calcite skeletons of trillions of marine organisms fell to the ocean floor, fossilized, and formed limestone.